```
!pip install transformers
!pip install torch
!pip install pandas
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/pyt
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/di
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/pyt
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/d
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.1
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/pytho
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import pandas as pd
```

```
model_name = "facebook/bart-large-cnn"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

def summarize_text(text, max_length=150, min_length=30, do_sample=False):
    inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        max_length=max_length,
        min_length=min_length,
        do_sample=do_sample,
        early_stopping=True,
        forced_bos_token_id=tokenizer.bos_token_id
    )
    summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
    return summary
```

```
Loading weights: 100%   511/511 [00:00<00:00, 892.30it/s, Materializing param=model.encoder.layers
```

```
data = pd.read_csv("summarization_dataset_lab18.csv")

print(data.head())
```

```
   ID           Text_Type                                                Text
0   1   Technical Article  Artificial Intelligence (AI) is transforming h...
1   2    News/Environment  Climate change has become one of the most pres...
2   3      Education Blog  Online education has grown rapidly over the pa...
3   4  Technology Article  The development of modern smartphones has chan...
4   5        Science News  Space exploration has fascinated humanity for ...
```

```
texts = data["Text"].tolist()
```

```
for text in texts:
    summary = summarize_text(text, max_length=120, min_length=40, do_sample=Fal

    print("ORIGINAL TEXT:\n", text)
    print("\nSUMMARY:\n", summary)
    print("\n----------------------------------------------------\n")
```

```
ORIGINAL TEXT:
 Artificial Intelligence (AI) is transforming healthcare by enabling faster diag

SUMMARY:
 Artificial Intelligence is transforming healthcare by enabling faster diagnosis

----------------------------------------------------

ORIGINAL TEXT:
```

```
 Climate change has become one of the most pressing global issues of the 21st ce

SUMMARY:
 Climate change has become one of the most pressing global issues of the 21st ce

--------------------------------------------------

ORIGINAL TEXT:
 Online education has grown rapidly over the past decade, especially after the C

SUMMARY:
 Online education has grown rapidly over the past decade, especially after the C

--------------------------------------------------

ORIGINAL TEXT:
 The development of modern smartphones has changed the way people communicate, w

SUMMARY:
 Smartphones combine multiple technologies including cameras, internet connectiv

--------------------------------------------------

ORIGINAL TEXT:
 Space exploration has fascinated humanity for centuries, and recent advancement

SUMMARY:
 Government space agencies and private companies are launching missions to explo

--------------------------------------------------

ORIGINAL TEXT:
 Customer service conversations are increasingly handled by AI-powered virtual a

SUMMARY:
 Customer service conversations are increasingly handled by AI-powered virtual a

--------------------------------------------------
```

```
summary_short = summarize_text(texts[0], max_length=50, min_length=20, do_sampl
print(summary_short)
```

```
Artificial Intelligence (AI) is transforming healthcare by enabling faster diagn
```

```
summary_medium = summarize_text(texts[0], max_length=100, min_length=40, do_sam
print(summary_medium)
```

```
Artificial Intelligence is transforming healthcare by enabling faster diagnosis,
```

```
summary_long = summarize_text(texts[0], max_length=150, min_length=70, do_
print(summary_long)
```

```
Artificial Intelligence (AI) is transforming healthcare by enabling faster diagn
```

```
long_text = texts[0] + texts[1]

summary = summarize_text(long_text, max_length=150, min_length=60, do_samp

print(summary)
```

```
Artificial Intelligence is transforming healthcare by enabling faster diagnosis,
```

Start coding or generate with AI.